1-Step vs 2-Step LLM Classification: Comparison

Date: 2026-07-31 **Purpose:** Compare the original single-pass Qwen→GPT/Sonnet/Opus pipeline (07_31 run) against the new two-step pipeline (step-1 simple Qwen pre-filter → step-2 GPT-4o/Sonnet/Opus with detailed rules), and additionally test whether including Qwen's step-1 flagging reason as context in the step-2 prompt materially affects results.

Pipeline Overview

1-Step Approach (07_31, `compare_bills_07_31_3model.py`)

- Qwen runs on 5,000 bills using the **full detailed prompt** (`8_llm_categorize_VT27_bills_prompt_07_31.md`)
- Bills Qwen flags go directly to GPT-4o, Claude Sonnet, and Claude Opus for second-pass verification, using the same detailed prompt
- **140 bills** passed Qwen's first filter
- Manual review triggered on GPT-4o vs Opus disagreements

2-Step Approach with Reason (07_31 step2, `compare_206bills_07_31_step2.py`)

- Qwen runs on 5,000 bills using a **simplified step-1 prompt** designed to minimize false negatives (looser criteria, casts a wider net)
- Bills Qwen flags go to GPT-4o, Claude Sonnet, and Claude Opus for second-pass verification using `prompt_step2_gpt.md`, which applies the full detailed rules **and passes Qwen's step-1 flagging reason** to the verifier as context
- **206 bills** passed the step-1 filter (66 more than the 1-step approach)

2-Step Approach without Reason (07_31 step2b, `compare_206bills_07_31_step2b.py`)

- Same simplified Qwen step-1 pre-filter; same 206-bill input pool
- Same three-model step-2 verification with `prompt_step2_gpt.md` and full detailed rules, but **Qwen's step-1 flagging reason is NOT passed** to the verifiers
- Isolates the effect of the reason context on model agreement

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Results Comparison

Flagging Counts

Metric	1-Step (140 bills)	2-Step w/ reason (206 bills)	2-Step no reason (206 bills)
Input pool (after Qwen/step-1)	140	206	206
GPT-4o flagged	104 (74.3%)	114 (55.3%)	80 (38.8%)
Sonnet flagged	128 (91.4%)	104 (50.5%)	114 (55.3%)
Opus flagged	124 (88.6%)	107 (51.9%)	103 (50.0%)
All-model agree	**107 / 140 (76.4%)**	**169 / 206 (82.0%)**	**156 / 206 (75.7%)**

GPT-4o vs Sonnet Disagreements

Metric	1-Step	2-Step w/ reason	2-Step no reason
GPT vs Sonnet disagree	27 / 140 **(19.3%)**	28 / 206 **(13.6%)**	40 / 206 **(19.4%)**
GPT=flag, Sonnet=reject	1	19	3
GPT=reject, Sonnet=flag	26	9	37

GPT-4o vs Opus Disagreements

Metric	1-Step	2-Step w/ reason	2-Step no reason
GPT vs Opus disagree	24 / 140 **(17.1%)**	17 / 206 **(8.3%)**	29 / 206 **(14.1%)**

Any-Model Disagreements (all 3 models)

Metric	1-Step	2-Step w/ reason	2-Step no reason
Any model disagrees	33 / 140 (23.6%)	37 / 206 (18.0%)	50 / 206 (24.3%)
All 3 confirm as qualifying	101	89	74
All 3 confirm as non-qualifying	6	80	82
Sent to manual review	24 → `manual_review_07_31_GPT_vs_Opus_24.xlsx`	17 → `manual_review_07_31_2step_with_reason_GPT_vs_Opus_17.xlsx`	29 → `manual_review_07_31_2step_wo_reason_GPT_vs_Opus_29.xlsx`